Marimuthu Mayavan

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Seeking a 6-month internship from February 2027



Education

EURECOM Sep 2025 – Present
Master of Science in Networks & Telecommunications – IoT track *Sophia Antipolis, France*
Coursework: IoT Protocols, Network Security, Mobile and Smartphone Security, Deep Learning

Bharathidasan University Jun 2017 – May 2023
Integrated Master of Technology (6-year programme) in Computer Science and Engineering *Tiruchirappalli, India*
Coursework: Computer Networks, Network Security, IoT, Machine Learning

Technical Skills

Languages: Python (PyTorch, NumPy), C, C++, Java, MATLAB
Networking & IoT: TCP/IP, LAN/Wi-Fi troubleshooting, switches and access points, MQTT, CoAP
Tools & Modelling: Linux, Windows, Git/GitHub, Jupyter, SysML v2, UML, MBSE, Ansys System Architecture Modeler
Spoken: English (fluent), Tamil (native), French (A1)

Projects

HAR-DeepLearning · Python, PyTorch, CNN/LSTM Apr 2026
- Built and compared 1D-CNN, LSTM and CNN-LSTM models for activity recognition on wearable IoT sensor data (UCI HAR, 30 people)
- Reached 94% accuracy with the LSTM on unseen people; still works without the gyroscope (93% → 82%)

PickErUp · C, Linux (ev3dev), Git Dec 2025 – Jan 2026
- Wrote the movement and rotation functions in C (motor control) for an autonomous LEGO EV3 robot that finds ores, picks them up with a claw and brings them back to its base
- Helped integrate the main control program using sonar, colour and gyro sensors with 4 motors on Debian Linux

PayPal "Pay in 4" Privacy Impact Assessment · GDPR, CNIL PIA, EU AI Act Oct 2026 – Present
- Analysing PayPal's privacy policy, assessing privacy risks with the CNIL PIA tool (acting as DPO) and classifying its automated creditworthiness system under the EU AI Act

Bank CCTV Theft Detection · C#/.NET, SQL Server 2023
- M.Tech final project: built an application that detects motion in restricted hours, captures faces (Haar cascade) and sends email and SMS alerts with the face image

Paddy Crop-Row Detection from UAV Images · MATLAB 2021
- Detected crop rows and failures in paddy fields from drone images (wavelet clustering, morphological segmentation)

Research Experience

EURECOM, *Graduate Researcher (Semester Project)* Sep 2026 – Present
- Modelling automotive system architectures in SysML v2 with Ansys System Architecture Modeler, applying model-based systems engineering (MBSE) to use cases provided by Ansys (200 h, supervised by Rabea Ameur-Boulifa)

EURECOM, *Graduate Researcher (Semester Project)* Feb 2026 – Jun 2026
- Designed a course on audiovisual deepfake detection (100 h, Prof. Jean-Luc Dugelay): reviewed research on audio-visual fusion, lip-sync detection, datasets (FakeAVCeleb, DFDC, LAV-DF) and metrics (AUC-ROC, EER)
- Built 5 course modules with quizzes and a narrated AI-generated video lecture (HeyGen)

Professional Experience

Orbion Pharmaceuticals, *L1 Network Engineer (apprenticeship)* Dec 2023 – Nov 2024
- Provided first-level network support for 80+ PCs: LAN/Wi-Fi connectivity, IP configuration, switches and access points
- Diagnosed faults with ping, traceroute and ipconfig; installed desktops and printers; escalated complex issues to L2

Foxconn Hon Hai Technology India, *IP Camera Network Support* Apr 2025 – Aug 2025
- Installed new IP cameras, configured their IP addresses and added them to the video management software (VMS)
- Diagnosed offline cameras with ping and restored connectivity on PoE switches (power checks, port restarts)
- Replaced faulty cameras and cables, crimped RJ45 cables and configured NVR/DVR recording settings
- Supported, with a team, a CCTV network of 3,000+ IP cameras on a large manufacturing site; logged incidents

Certification

- Diploma in Computer Programming (C, C++, Python), TCDS Computer Education, 2023 – Distinction (91/100)